EXPERIMENT NO 7

CODE:

Import Libraries

```
import numpy as np import pandas as pd

import tensorflow as tf

from sklearn.model_selection import

train_test_split

from sklearn.preprocessing import

LabelEncoder, StandardScale
```

2. Load Dataset

```
df =

pd.read_csv('/content/Churn_Modelling.csv')

# Display dataset print(df.head()

# 3. Data Preprocessing

# Drop unnecessary columns

df = df.drop(['RowNumber', 'CustomerId',

'Surname'], axis=1)
```

```
# Encode categorical variables label_encoder

= LabelEncoder()df['Gender'] =

label_encoder.fit_transform(df['Gender'])

# One-hot encoding for Geography

df = pd.get_dummies(df,

columns=['Geography'], drop_first=True)

# Split features and target

X = df.drop('Exited', axis=1) y = df['Exited'

# 4. Train-Test Split

X_train, X_test, y_train, y_test =

train_test_split( X, y, test_size=0.2,

random_state=42

# 5. Feature Scaling

sc = StandardScaler()

X_train = sc.fit_transform(X_train) X_test =

sc.transform(X_test)

# 6. Build ANN Model

model = tf.keras.models.Sequential()



# Input + Hidden Layers
```

```
    model.add(tf.keras.layers.Dense(units=16,
    activation='relu',
    input_dim=X_train.shape[1]))
    model.add(tf.keras.layers.Dense(units=8,
    activation='relu'))
   # Output Layer
   model.add(tf.keras.layers.Dense(units=1,
activation='sigmoid')) #
================================
    # 7. Compile Model
model.compile( optimizer='adam',
loss='binary_crossentropy',
metrics=['accuracy']
    )
# 8. Train Model
history = model.fit( X_train, y_train,
batch_size=32, epochs=50,
validation_split=0.2
    )
# 9. Evaluate Model
```

```
loss, accuracy = model.evaluate(X_test,

y_test) print("Test Accuracy:", accuracy)

# 10. Predictions

y_pred = model.predict(X_test) y_pred =

(y_pred > 0.5)

# 11. Confusion Matrix

from sklearn.metrics import confusion_matrix

cm = confusion_matrix(y_test, y_pred)

print("Confusion Matrix:\n", cm)

    plt.plot(history.history['accuracy'],

    label='Training Accuracy')

    plt.plot(history.history['val_accuracy'],

    label='Validation Accuracy') plt.legend()

    plt.title("Model Accuracy") plt.show()
```

OUTPUT:

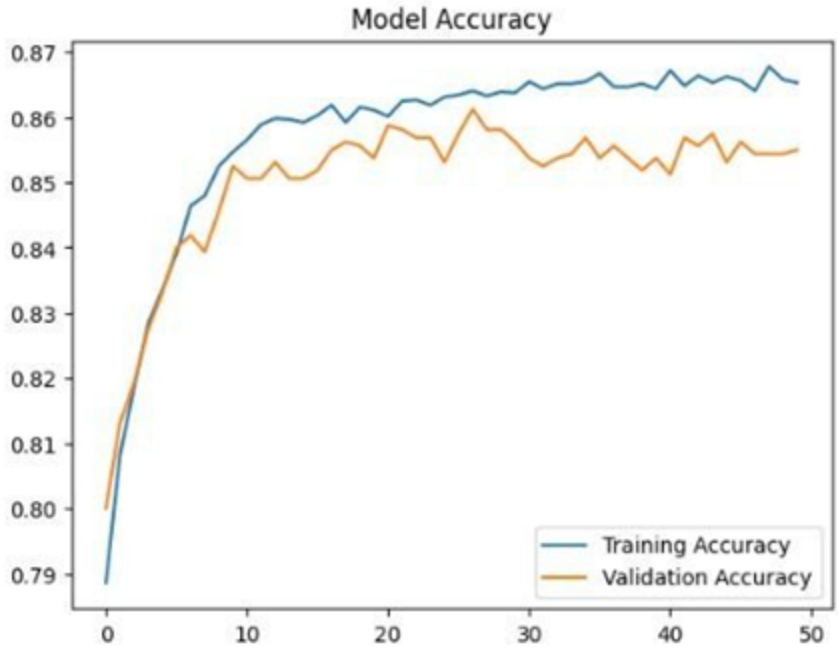
```
...    RowNumber  CustomerId   Surname  CreditScore Geography  Gender  Age  \
   0          1    15634602  Hargrave          619    France  Female   42
   1          2    15647311      Hill          608     Spain  Female   41
   2          3    15619304      Onio          502    France  Female   42
   3          4    15701354      Boni          699    France  Female   39
   4          5    15737888  Mitchell          850     Spain  Female   43

      Tenure    Balance  NumOfProducts  HasCrCard  IsActiveMember  \
   0       2       0.00              1          1               1
   1       1   83807.86              1          0               1
   2       8  159660.80              3          1               0
   3       1       0.00              2          0               0
   4       2  125510.82              1          1               1

      EstimatedSalary  Exited
   0        101348.88       1
   1        112542.58       0
   2        113931.57       1
   3         93826.63       0
   4         79084.10       0
   Epoch 1/50
   /usr/local/lib/python3.12/dist-packages/keras/src/layers/core/dense.py:106: UserWarning: Do not pass an `input_shape`/`input_dim` argument to
     super().__init__(activity_regularizer=activity_regularizer, **kwargs)
   200/200 ━━━━━━━━━━━━━━━━━━━━ 3s 5ms/step - accuracy: 0.7886 - loss: 0.4907 - val_accuracy: 0.8000 - val_loss: 0.4517
   Epoch 2/50
   200/200 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - accuracy: 0.8083 - loss: 0.4383 - val_accuracy: 0.8131 - val_loss: 0.4327
   Epoch 3/50
   200/200 ━━━━━━━━━━━━━━━━━━━━ 1s 2ms/step - accuracy: 0.8186 - loss: 0.4225 - val_accuracy: 0.8194 - val_loss: 0.4240
   Epoch 4/50
   200/200 ━━━━━━━━━━━━━━━━━━━━ 1s 2ms/step - accuracy: 0.8284 - loss: 0.4096 - val_accuracy: 0.8275 - val_loss: 0.4104
   Epoch 5/50
   200/200 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - accuracy: 0.8336 - loss: 0.3977 - val_accuracy: 0.8331 - val_loss: 0.4000
   Epoch 6/50
   200/200 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - accuracy: 0.8391 - loss: 0.3859 - val_accuracy: 0.8400 - val_loss: 0.3901
   Epoch 7/50
   200/200 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - accuracy: 0.8464 - loss: 0.3754 - val_accuracy: 0.8419 - val_loss: 0.3799
   Epoch 8/50
   200/200 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - accuracy: 0.8480 - loss: 0.3665 - val_accuracy: 0.8394 - val_loss: 0.3717
```

```
200/200 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - accuracy: 0.8662 - loss: 0.3234 - val_accuracy: 0.8531 - val_loss: 0.3469
Epoch 46/50
200/200 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - accuracy: 0.8656 - loss: 0.3226 - val_accuracy: 0.8562 - val_loss: 0.3479
Epoch 47/50
200/200 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - accuracy: 0.8641 - loss: 0.3222 - val_accuracy: 0.8544 - val_loss: 0.3496
Epoch 48/50
200/200 ━━━━━━━━━━━━━━━━━━━━ 1s 2ms/step - accuracy: 0.8678 - loss: 0.3223 - val_accuracy: 0.8544 - val_loss: 0.3487
Epoch 49/50
200/200 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - accuracy: 0.8658 - loss: 0.3219 - val_accuracy: 0.8544 - val_loss: 0.3484
Epoch 50/50
200/200 ━━━━━━━━━━━━━━━━━━━━ 1s 2ms/step - accuracy: 0.8653 - loss: 0.3217 - val_accuracy: 0.8550 - val_loss: 0.3464
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - accuracy: 0.8610 - loss: 0.3403
Test Accuracy: 0.8610000014305115
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step
Confusion Matrix:
 [[1530   77]
 [ 201  192]]
```

...


Model Accuracy
0.87
0.86
0.85
0.84
0.83
0.82
0.81
0.80
0.79
0
10
20
30
40
50
Training Accuracy
Validation Accuracy